The Paper Project

— Sahu, S

Paper is the truest form of a document, physical, permanent, and universally readable. 'The Paper Project' brings together Markdown and LaTeX with diagrams, formulae, and rich formatting and renders the result as a PDF.

Overview

`paper` takes a Markdown file and renders it as a styled PDF via a Docker container. It handles:

- Custom LaTeX template with section-aware headers and page numbers
- Emoji rendering (unicode → SVG via twemoji)
- Mermaid diagrams (rendered via headless Chrome)
- Toggles for page numbers and section page breaks

Requirements

- Docker

Installation

Clone the repo and build the Docker image:

```
git clone <repo-url> $HOME/paper
cd $HOME/paper
docker build -t paper:latest .
```

Then symlink the `paper` script into a directory on your `$PATH`:

```
ln -sf $HOME/paper/paper $HOME/.local/bin/paper
```

Usage

```
paper <file.md> [options]
```

Option	Description
`--pages=yes`	Show page numbers (default)
`--pages=no`	Hide page numbers
`--continuous`	No page break between sections
`-o`, `--output <file.pdf>`	Write PDF to this path (default: temp dir, opened automatically)
`-v`, `--verbose`	Print commands as they run

After the PDF is generated it opens automatically. You are then prompted whether to keep or delete it.

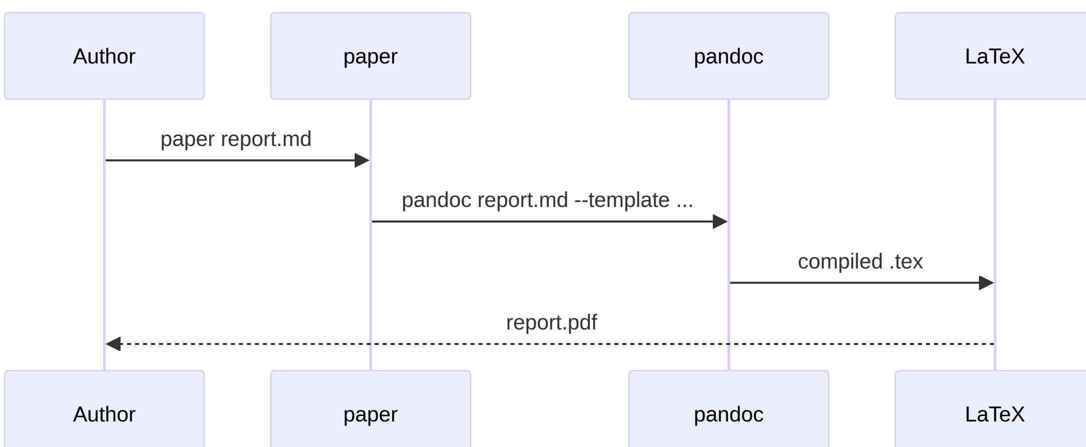
Examples

```
# Basic
paper report.md

# No page numbers
paper report.md --pages=no

# No section breaks, no page numbers
paper report.md --continuous --pages=no

# Save PDF to a specific path
paper report.md -o ~/Documents/report.pdf
```

Mermaid diagrams

Mermaid code blocks are rendered as diagrams in the PDF.

````
```mermaid
graph TD
    A[Write Markdown] --> B[Run paper]
    B --> C{Success?}
    C -- Yes --> D[Open PDF]
    C -- No --> E[Fix errors]
    E --> B
```
````



Front matter variables

Set these in your document's YAML front matter to control layout:

```
---
title: My Report
author: Your Name
date: 2026-06-13
continuous-pages: true   # no page breaks between sections
no-page-numbers: true    # hide page numbers
---
```

Acknowledgements

`src/emoji-filter.js` is based on work by Miguel Angelo, originally licensed under the Apache License 2.0. It converts unicode emoji in Markdown to SVG images (via twemoji or noto-emoji) so they render correctly in the PDF output.

`src/pandoc-emoji-filter.js` is derived from masbicudo/Pandoc-Emojis-Filter, modified to suit this project. It preprocesses Markdown by replacing unicode emoji with local SVG image references before the document is passed to pandoc.